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ADAPTIVE AND EXPLAINABLE HUMAN-ROBOT COLLABORATION FOR INTELLIGENT SORTING SYSTEMS

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Collaborative Texture Recognition



? Problem

No existing system reliably sorts glass, plastic, rubber, and metal in real time using vision, sensors, and human feedback.

Main Purpose

Enable texture based object recognition using sensors and vision

Solution



Camera
(Visual Data)

DINOv2 (ViT)



Ultrasonic sensor
(Distance Signals)

Sensor Feature
Extraction

Multimodal
Classification Model

Material Classification



Target Users

Children and differently abled students to identify and interact with textures

Real World Impact

Promotes inclusive learning and independence through vision and sensors



Adaptive Pattern Recognition via Human Behavior

Target Users

Differently abled children with limited physical movement

Main Purpose

allows children to learn patterns and give verbal exercises

Real World Impact

Empowering independence through accessible interaction



Overview



Real Time Object Detection



Human Pattern Detection



Robot Sorting Strategy



Sort



Human voice Detection



Human-Led Selection Through Gaze-Guide Robotic Assistance



Target Users

Differently abled children with limited physical movement

Main Purpose

allows children to select objects without using physical controls

Real World Impact

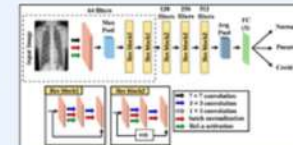
Empowering independence through accessible interaction



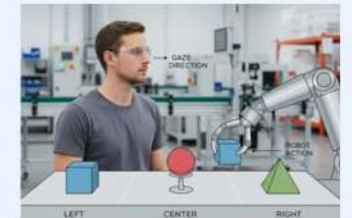
Overview



DNN/Resnet18



Region Mapping
With Gaze Direction



Our solution uses a standard RGB camera to capture human gaze and interpret intent without any wearable or intrusive hardware.

Integrating Explainable AI and Adaptive Recommendations for Human-Centred Robotic Sorting



Target Users

Differently abled children with limited physical movement

Main Purpose

allows children to select objects without using physical controls

Real World Impact

Empowering independence through accessible interaction



Risk Mitigation

- Evaluation Metrics
- Data Set Creation over real incidents
- Smart Recommendation providing



Solution



Real Time Object Detection



Generate Recommendations



Explain



Sort